

Exercise 2: Expectation and Variance

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Problem 1 (LOTUS for a Continuous Random Variable):

Let X be a continuous random variable with probability density function (PDF) $f_X(x)$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued, differentiable and **strictly increasing** function. Prove the Law of the Unconscious Statistician (LOTUS), which states that the expected value of the random variable $Y = g(X)$ is given by:

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x) dx.$$

Remark: The result holds for any function g , but its proof becomes much more involved.

Problem 2 (Properties of Expected Value):

Prove the following properties of the expected value operator $\mathbb{E}[\cdot]$.

- (a) The expected value of a constant is itself: $\mathbb{E}[c] = c$.
- (b) Linearity of Expectation: For random variables X, Y and scalars a, b , it holds that $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. *Hint: Use LOTUS for the random vector (X, Y) , the relationship between joint and marginal distributions and Fubini's theorem.*
- (c) Monotonicity of Expectation: If X and Y are two random variables such that $X(\omega) \geq Y(\omega)$ for all outcomes ω , then $\mathbb{E}[X] \geq \mathbb{E}[Y]$.
- (d) Expectation of a Product: If the random variables $X_1, \dots, X_n \in \mathbb{R}$ are **mutually independent**, then $\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n]$.

Hint: Use LOTUS for a random vector, Fubini's theorem and that if $X_1, \dots, X_n$ are mutually independent then $f_{X_1, \dots, X_n}(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$.

Problem 3 (Zero Variance):

Let X be a random variable with a finite expected value. Show that $\text{Var}(X) = 0$ if and only if X is a constant almost surely, i.e. it takes a constant value with probability 1.

Problem 4 (Variance of the Exponential Distribution):

The **Exponential Distribution** is often used to model the time until an event occurs in a Poisson process. Its probability density function (PDF) is given by:

$$f(x) = \lambda e^{-\lambda x}, \quad \text{for } x \geq 0,$$

where $\lambda > 0$ is the rate parameter. Compute the variance, $\text{Var}(X)$, of the exponential distribution.

Problem 5 (Variance of a Sum):

Show that for two random variables X and Y , the variance of their sum is given by:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

Then, for a collection of n mutually independent random variables $X_1, \dots, X_n$, show that:

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Hint: Use that if $X_1, \dots, X_n$ are mutually independent then $f_{X_1, \dots, X_n}(x_1, \dots, x_n) = f_{X_1}(x_1) \dots f_{X_n}(x_n)$ and Fubini's theorem.

Problem 6 (Covariance for a Joint Continuous Distribution):

Let the continuous random variables X and Y have a joint PDF given by:

$$f(x, y) = \begin{cases} c & \text{if } 0 \leq y \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

This is a uniform distribution over the triangle defined by the lines $y = 0$, $x=1$, and $y=x$.

Compute the covariance, (X, Y) , by following these steps:

- (a) Find the value of the normalization constant c .
- (b) Find the marginal PDFs, $f_X(x)$ and $f_Y(y)$.
- (c) Compute the expected values, $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.
- (d) Compute the expectation of the product, $\mathbb{E}[XY]$.
- (e) Use the above results to find $\text{Cov}(X, Y)$.